Master's Program in Applied Economics — IDEEP

University of Lille

Copper Exploration Potential in Arizona and Nevada: Spatial Analysis and Exploratory Scoring Based on Open Data

Exploratory Analysis Report

Lucas Lokumu

Academic Year 2025-2026

Table of Contents

Introduction

This report presents an exploratory analysis of copper exploration potential in Arizona and Nevada, two U.S. states with a long tradition of mining activity. The study relies exclusively on open data, in particular the MRDS (Mineral Resources Data System) database published by the USGS (United States Geological Survey), together with complementary administrative and cartographic reference layers.

The objective is not to assess the economic profitability of mining sites, but to propose a reproducible method for ranking a large number of listed sites and, based on explicit criteria, to bring out a subset of sites and territories considered priorities for further analysis. The work combines two complementary tools: QGIS, for spatial processing and mapping, and Python, for automating descriptive statistics and producing charts.

The research question can be formulated as follows: how can open mining data be used to rank copper sites in Arizona and Nevada, and to identify priority areas for further exploration analysis? This question requires building a processing chain from raw, heterogeneous, and voluminous data to a synthetic territorial reading, while maintaining the methodological caution required for this type of exploratory exercise.

The report is structured in several parts. After presenting the context and the data used, the methodology is detailed step by step, from importing the MRDS data to constructing a historical mining context indicator. The results are then presented and discussed, followed by a section on the limitations of the exercise and possible extensions.

Context and Interest of Copper

Copper occupies a particular place among mineral raw materials. It is used extensively in the electrification of uses, electricity distribution networks, renewable energy production infrastructure, and, more broadly, in a large share of industrial equipment. This central role makes it a resource closely monitored in discussions on the energy transition, alongside other metals such as lithium, nickel, or cobalt.

Arizona and Nevada are among the most historically active U.S. states in terms of mining activity. Their subsurface has been the subject of exploration and mining campaigns for more than a century, leaving a substantial trace in public geological databases. This historical depth is an asset for this type of analysis: it provides a sufficient volume of data to build statistically significant indicators, while covering territories with contrasting geological characteristics.

The choice of these two states is therefore based on three elements: their recognized geological potential, the availability of open data of sufficient quality for this type of work, and the value of comparing two neighboring yet distinct territories. This study specifically seeks to turn a raw, descriptive database into a territorial reading tool that can be more directly used to guide exploratory thinking.

Data Used

The analysis relies on four main sources, chosen for their public nature and their homogeneous coverage of the study area.

USGS ScienceBase — entry point for the copper project for Arizona and Nevada, bringing together the datasets used in this study

https://www.sciencebase.gov/catalog/item/64dfc0efd34e5f6cd553c265

USGS MRDS (Mineral Resources Data System) — database listing known occurrences, mines, prospects, and mineral sites across the United States; this is the central source for this work

https://mrdata.usgs.gov/mrds/

U.S. Census Bureau — Cartographic Boundary Files — provides the administrative boundaries of U.S. states, used to delineate the study area

https://www.census.gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html

OpenStreetMap — used as a basemap in QGIS, via an XYZ tile stream, to give a readable geographic reference to the maps produced.

In addition to these sources, the USGS TopoMineSymbols Consolidate layer was used, which records historical mining traces (topographic symbols indicating former mines or workings). This layer is not used for its geological completeness, but as a context indicator: it makes it possible to measure the density of past mining activity around the identified copper sites.

All processing was organized in a working folder named "Copper Exploration." Intermediate and final files (filtered, merged, and classified layers, as well as the complete QGIS project) are documented in the methodological appendix rather than detailed in the body of the text, in order to keep the report easy to read.

Methodology

The methodology follows a multi-step processing chain, from importing raw data to building synthetic indicators. Each step is described below, together with the choices made and their justification.

MRDS Data Preparation

MRDS data were downloaded separately for Arizona and Nevada, then imported into QGIS in shapefile / GeoPackage format. The initial volume is substantial: 12,951 records for Arizona and 15,444 for Nevada, across all mineral commodity types combined. This large volume illustrates the richness of the MRDS database, but also the need for filtering to isolate information relevant to copper.

Filtering Copper-Related Sites

Filtering of copper sites relies on three commodity fields present in the MRDS database: commod1, commod2, and commod3. These fields respectively indicate the primary commodity and the secondary commodities associated with each site. The selection expression used in QGIS is as follows:

```
"commod1" LIKE '%Copper%' OR "commod2" LIKE '%Copper%' OR "commod3" LIKE
'%Copper%'
```

Using all three fields is necessary because copper can appear as the primary commodity of a site, but also as a by-product of a polymetallic deposit. Limiting the filter to the commod1 field alone would have excluded sites where copper is present but secondary, which would have biased the final database toward pure copper deposits only.

Merging Arizona and Nevada

The filtered layers for Arizona and Nevada were then merged to form a common database, MRDS_copper_AZ_NV.gpkg. This merge allows the two states to be processed within the same spatial and statistical framework, while retaining the ability to distinguish between them afterward using the state-of-origin field.

After merging, the database brings together 8,789 copper-related sites, unevenly distributed between the two states: 5,242 sites in Arizona and 3,536 sites in Nevada. A few marginal records, associated with border areas, may remain in the database, but the analysis remains focused on the territory of these two states.

Building an Exploratory Scoring System

A scoring system was built to rank the 8,789 recorded copper sites. This is not an expert geological model, but a ranking tool based on criteria available within the MRDS database itself. The overall score combines several dimensions:

- the development status of the site (dev_stat), which distinguishes, for example, between past producers, occurrences, or prospects;

- the quality of the MRDS record (score field), which reflects the reliability and completeness of the site's record;
- the deposit type (dep_type), when available;
- a total score aggregating these criteria;
- a final potential class, summarizing the total score into three categories: Low, Medium, High.

The resulting distribution is as follows: 4,902 sites in the Low class, 3,740 sites in the Medium class, and 147 sites in the High class. This result deserves comment: out of 8,789 recorded sites, only 147 — a little under 2% — reach the most favorable class. This selective nature is intentional. The objective of the scoring is not to validate all known sites, but to bring out a limited group of priority sites on which a more detailed analysis could then focus.

Spatial Analysis in QGIS

Beyond the attribute-based scoring, a spatial dimension was added to the analysis through the cross-referencing of high-potential sites with historical mining traces. This step is detailed in the results section, as it constitutes one of the main contributions of the study. On the mapping side, QGIS was also used to produce two summary maps, presented further below, combining an OpenStreetMap basemap, the administrative boundaries from the Census Bureau, and the layers of classified sites.

Python Automation

A Python script, named copper_analysis.py and placed in a scripts folder, was developed to make the analysis reproducible. It reads directly from the MRDS_copper_AZ_NV.gpkg and high_potential_10km_mining_points_count.gpkg layers and automatically produces tables in CSV format, saved in outputs/tables, as well as charts in PNG format, saved in outputs/figures.

The tables produced cover the distribution by potential class, the distribution of high-potential sites by state, by county, and by mining context, as well as a final priority table by county. The associated charts present these same dimensions in visual form. This script relies on the pandas, geopandas, and matplotlib libraries, commonly used for geospatial data processing and descriptive statistics in Python.

The Python processing complements the work carried out in QGIS. QGIS was used for spatial processing, mapping, and geographic joins, while Python was used to automate descriptive statistics, export tables, and generate the report's charts. This combination strengthens the reproducibility of the analysis: the entire chain, from raw data to final chart, can be re-run from the same source files.

Results

Distribution of Potential Classes

The distribution of the 8,789 copper sites by exploration potential class is summarized in the following table.

Potential Class	Number of Sites
Low	4,902
Medium	3,740
High	147

The small number of sites classified as high potential confirms the selective nature of the exploratory scoring. Most sites remain classified as Low or Medium, which avoids overinterpreting the MRDS database and encourages focusing attention on a smaller but better-characterized subset of sites.

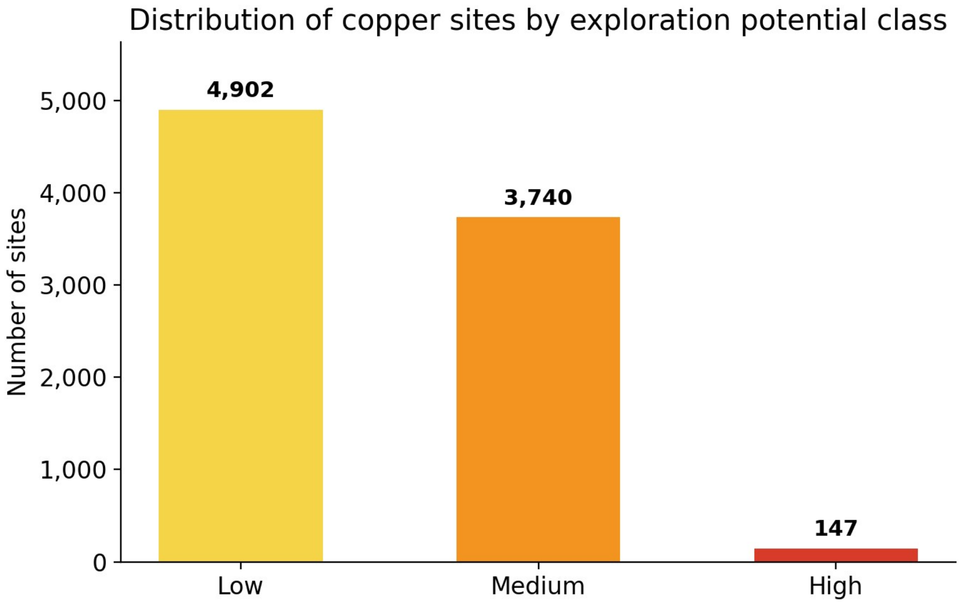


Figure 1 — Distribution of copper sites by exploration potential class.

High-Potential Sites by State

State	High-Potential Sites
Nevada	77
Arizona	70

The distribution of high-potential sites between the two states is relatively balanced, with a slight edge for Nevada. This result confirms that the analysis is not focused on a single territory, but highlights exploratory potential spread across the entire study area.

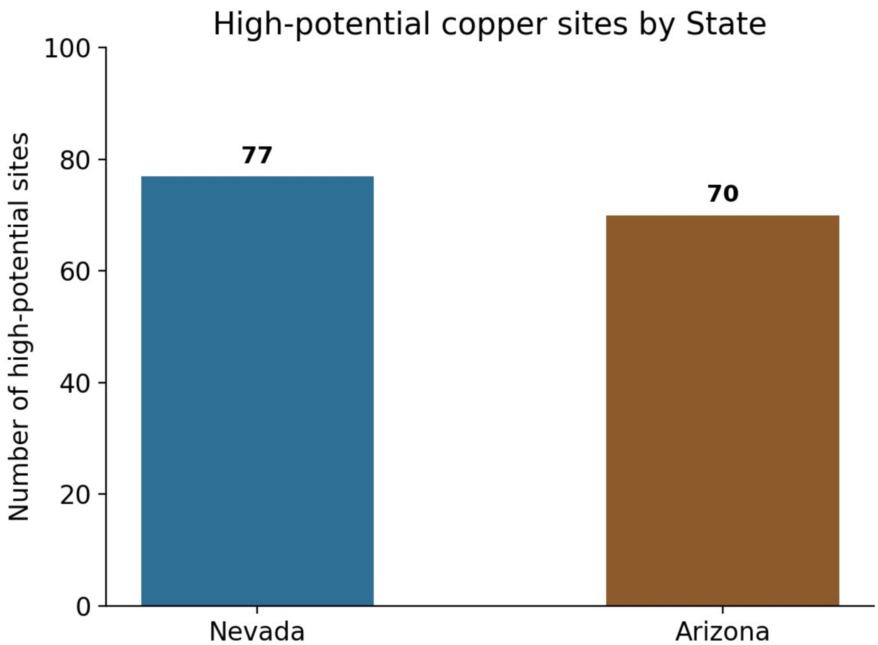


Figure 2 — High-potential copper sites, by State.

Analysis by Development Status and Deposit Type

The dev_stat field records the development status of each site. Its distribution across all 8,789 copper sites is as follows:

Status (dev_stat)	Number of Sites
Past Producer	4,604
Occurrence	1,727
Prospect	1,124
Producer	905
Unknown	371
Plant	58

A large share of the sites correspond to former producers or simple occurrences. This underscores the historical significance of the territory, but also calls for a degree of caution: a site that produced in the past is not necessarily economically relevant today, in particular due to technological, regulatory, and market developments that have occurred since it was worked.

The dep_type field, which describes the geological deposit type, is very heterogeneous and often missing, as shown in the following table.

Deposit Type (dep_type)	Number of Sites
Not recorded (NULL)	7,033
Vein	1,075
Replacement	241
Unknown	90
Contact metamorphic / metasomatic	≈ 55–56
Porphyry Cu	26
Porphyry Cu-Mo	11

The dep_type field is a useful piece of information when available, but its high rate of missing values limits its scope. On its own, it does not allow conclusions to be drawn about the economic potential of the sites concerned.

Quality of MRDS Records

The quality of the MRDS records, measured by the score field, breaks down as follows:

Quality (score)	Number of Sites
A	142
B	2,240
C	1,351
D	4,918
E	138

A large share of the records fall into category D, which is an important limitation of the analysis. The MRDS database remains useful for identifying large-scale spatial trends, but it does not replace detailed, site-by-site verified geological or economic data.

Cross-Referencing with Historical Mining Traces

Beyond the attribute-based scoring, the analysis sought to strengthen the spatial interpretation of the 147 high-potential sites. A dedicated layer of historical mining traces was first created (mining_features_points_AZ_NV_clean.gpkg) from the USGS TopoMineSymbols layer. A 10-kilometer buffer was then built around each of the 147 high-potential sites (buffer_10km_high_potential_copper_AZ_NV.gpkg).

A summary spatial join was then carried out, using the buffer layer as the input layer and the mining traces layer as the comparison layer, with an intersection predicate and a count-type summary. The result, stored in high_potential_10km_mining_points_count.gpkg, includes a fid_count field that measures, for each high-potential site, the number of historical mining traces present within a 10-kilometer radius.

A qualitative variable, mining_context, was then built from this field, using the following expression:

```
CASE WHEN "fid_count" >= 300 THEN 'Very strong' WHEN "fid_count" >= 100 THEN
'Strong' WHEN "fid_count" >= 30 THEN 'Medium' ELSE 'Low' END
```

Mining Context (10 km radius)	Number of Sites
Very strong	63
Strong	52
Medium	24
Low	8

Among the 147 copper sites classified as high potential, 63 are located in a very strong mining context and 52 in a strong context. This means that most priority sites are not isolated: they are located in territories already marked by historical mining activity. This result reinforces their interest for a more in-depth exploratory analysis, without allowing any direct conclusion about their profitability.

In total, 115 of the 147 high-potential sites, or approximately 78% of them, are located in a strong or very strong mining context. This result is one of the most significant contributions of the spatial analysis carried out in this work.

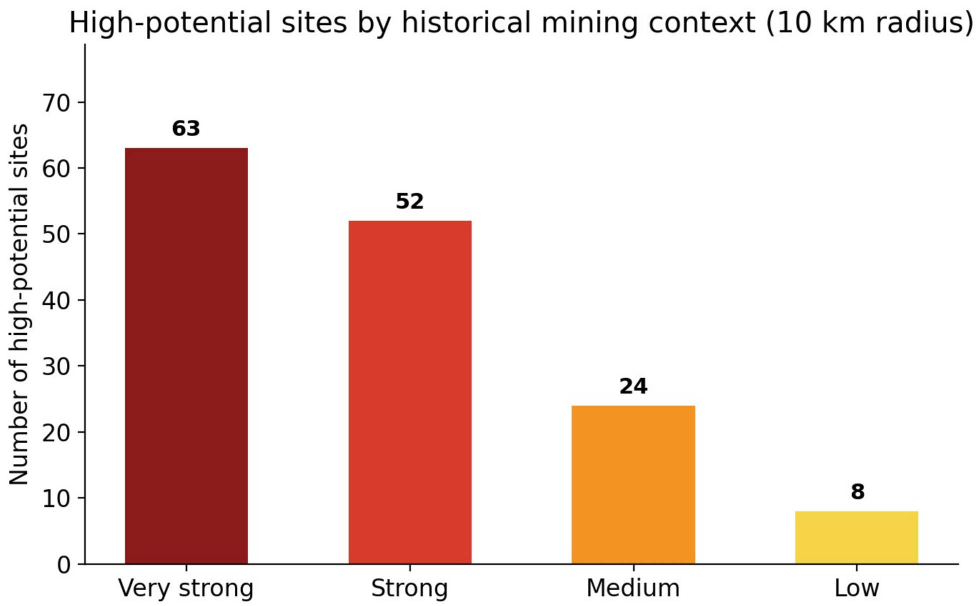
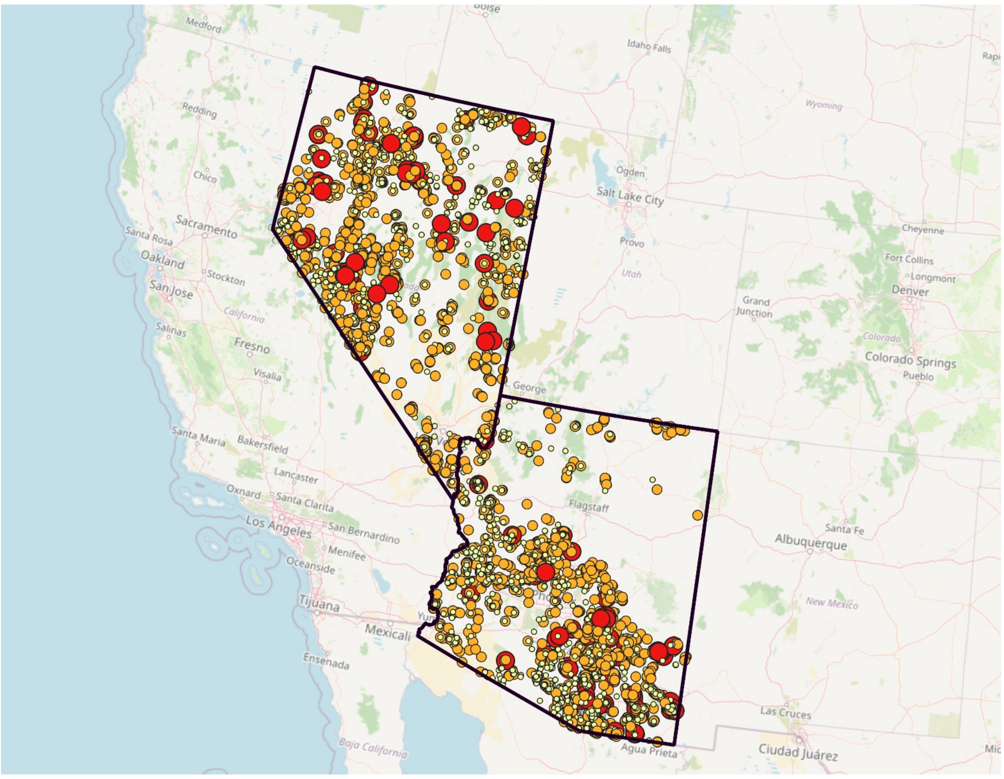


Figure 3 — Distribution of the 147 high-potential sites by historical mining context within a 10 km radius.

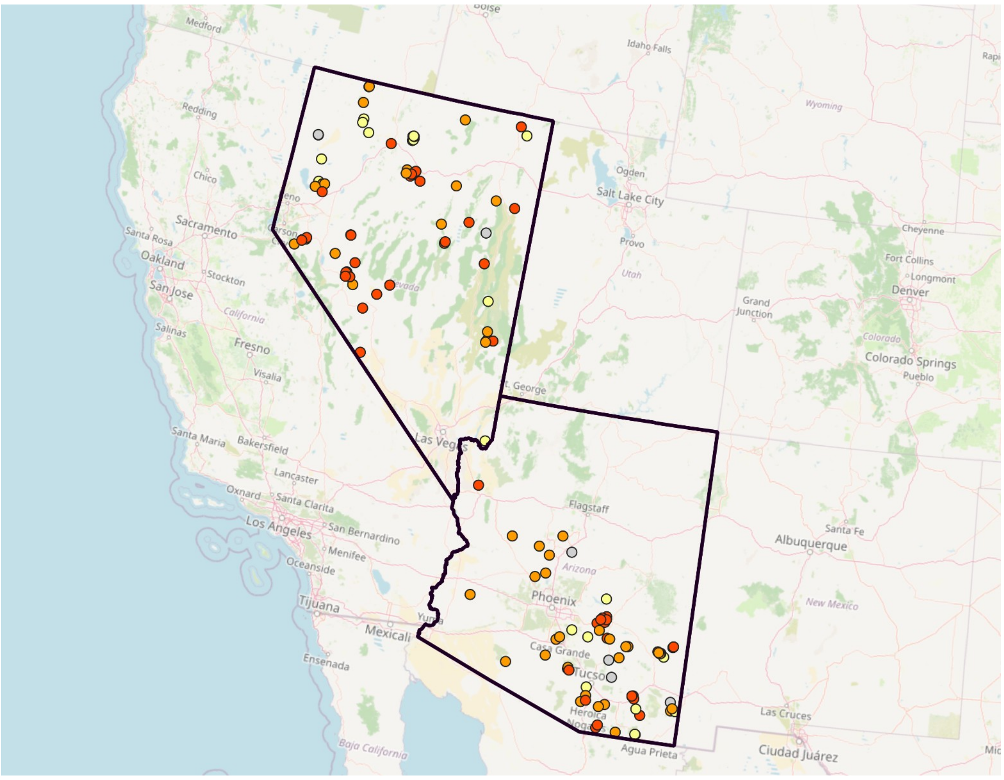
Mapping

Two maps were produced to illustrate the results. The first, titled "Copper Site Potential in Arizona and Nevada" (Carte_1_potentiel_cuivre_AZ_NV.png), overlays an OpenStreetMap basemap, the boundaries of the two states from the Census cb_2018_us_state_500k layer, and all copper sites classified by potential class, using a color scale ranging from yellow (Low) to orange (Medium) to red (High), with state boundaries shown in dark purple. This map offers an overall view of site distribution and shows a more pronounced concentration in southern Arizona as well as in several Nevada districts.



Map 1 — Copper site potential in Arizona and Nevada, classified by exploration potential class.

The second map, "Priority Copper Sites and Historical Mining Context within 10 km" (Carte_2_sites_prioritaires_contexte_minier_10km_AZ_NV.png), focuses exclusively on the 147 high-potential sites, classified by their mining context. More analytical than the first, it makes it possible to visually identify sites located in historically dense mining environments, consistent with the figures presented above.



Map 2 — Priority copper sites and historical mining context within 10 km, Arizona and Nevada.

Priority Counties

Aggregating high-potential sites at the county level makes it possible to move from a point-based reading to a territorial one. The following table presents the counties with the largest number of priority sites.

County	State	High-Potential Sites
Cochise	Arizona	16
Humboldt	Nevada	13
Pima	Arizona	12
Gila	Arizona	11
Mineral	Nevada	11
Lander	Nevada	11
Pinal	Arizona	10
Graham	Arizona	8
Yavapai	Arizona	6
Elko	Nevada	6
Pershing	Nevada	6
Lincoln	Nevada	5

The counties of Cochise, Pima, Gila, and Pinal structure the potential on the Arizona side, while Humboldt, Mineral, and Lander stand out as important areas on the Nevada side. These territories combine both a significant presence of high-potential sites and a notable density of historical mining traces. It should be clarified that this ranking does not mean that these counties would be the most profitable: it only indicates that, according to the criteria used in this study, they appear as priority areas for further analysis.

The following chart makes it possible to move from a site-level reading to a territorial one, indicating the counties where further analysis could be prioritized.

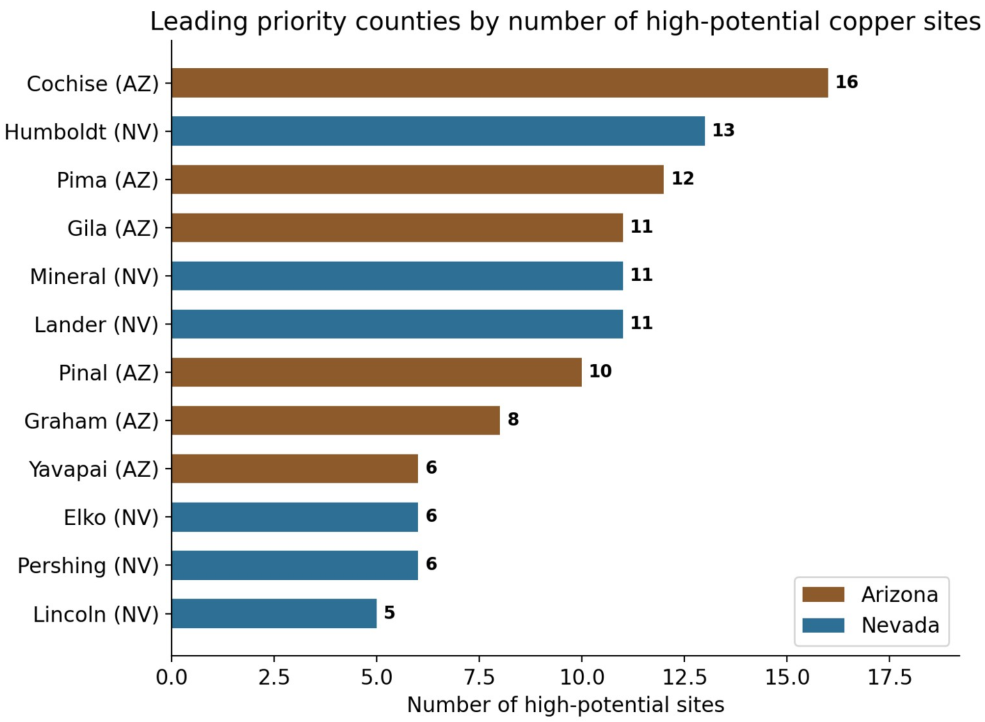


Figure 4 — Leading priority counties by number of high-potential copper sites.

Discussion

The results of this study must be interpreted with a limited and clearly defined scope. The exploratory scoring built from the fields available in MRDS does not constitute an expert geological assessment: it is a relative ranking tool, allowing a subset of sites with more favorable attribute characteristics, according to the criteria used, to be distinguished within a database of several thousand sites.

The territorial value of the analysis lies mainly in the cross-referencing between the attribute-based scoring and the historical mining context. The fact that 115 of the 147 high-potential sites are located in a strong or very strong mining environment suggests spatial consistency between the two approaches: sites deemed favorable by the scoring tend to be located in areas where historical mining activity has already been significant. This convergence strengthens the credibility of the ranking, without making it proof of economic potential.

Finally, the county-level reading makes it possible to propose a broader territorial ranking than the list of individual sites alone. It offers a relevant entry point for guiding further research, for example with the geological surveys of the states concerned or in the technical literature available on these counties. It should nonetheless be kept in mind that this ranking remains based on descriptive and historical data, and calls for interpretive caution, discussed in the following section.

Limitations

Several limitations must be highlighted to correctly situate the scope of this work.

The first concerns the heterogeneous quality of MRDS records. As noted above, nearly 56% of the copper sites in the database are classified as quality D, meaning their descriptive record is incomplete or unreliable. This variable quality level directly affects the robustness of the scoring built from these fields.

The second limitation relates to missing data, in particular the dep_type field, which is not recorded for 7,033 of the 8,789 copper sites. This lack of information on deposit type reduces the scoring's ability to incorporate a genuinely geological dimension, and requires greater reliance on status and record-quality criteria.

The third limitation, likely the most significant, concerns the total absence of economic data. MRDS data do not provide homogeneous information allowing a serious estimate of CAPEX, OPEX, copper grades, estimated resources or reserves, logistics costs, access to energy, environmental constraints, permits in force, available processing infrastructure, or the exact stage of progress of each project.

MRDS data make it possible to identify areas of geological and historical interest, but they do not make it possible to directly estimate the economic profitability of sites. A complete investment analysis would require additional data on grades, resources, CAPEX, OPEX, logistics costs, environmental constraints, and project development stage.

This limitation is essential to keep in mind: it clearly distinguishes the exploratory potential highlighted in this report from any estimate of economic profitability, which would fall under a mining feasibility study in the strict sense, outside the scope of this work.

Possible Extensions

Several extensions could enrich this exploratory analysis. On the infrastructure side, it would be useful to incorporate roads and distances to existing infrastructure, as well as cities, logistics hubs, or rail networks, in order to assess the accessibility of the identified sites. Adding power lines and available energy sources would also make it possible to indirectly approach potential operating costs.

On the environmental and regulatory side, cross-referencing with environmental zoning or land status data would provide essential additional insight before any operational step. Incorporating technical reports published by mining companies, where they exist for certain sites, would make it possible to spot-check the relevance of the ranking obtained.

Finally, on the methodological side, incorporating grade, resource, and reserve data, building a CAPEX/OPEX proxy from public comparables, or developing a broader territorial attractiveness index would be natural developments of this work. Comparing Arizona and Nevada with other mining states or provinces, as well as developing an interactive dashboard, would ultimately make it possible to generalize this approach and share it more widely.

Conclusion

This study made it possible to turn an open, raw, and voluminous database into a territorial reading tool for copper exploration potential in Arizona and Nevada. In total, 8,789 copper sites were recorded in the final database, of which 147 were classified as high potential according to the criteria used, split between 77 sites in Nevada and 70 in Arizona.

Cross-referencing with historical mining traces showed that 115 of these 147 sites, or approximately 78% of them, are located in a strong or very strong mining context, which reinforces the spatial consistency of the ranking obtained. Certain counties finally stand out as priority areas for further analysis, notably Cochise, Humboldt, Pima, Gila, Mineral, and Lander.

These results in no way prove the economic profitability of the identified sites. They only indicate, using an explicit and reproducible method, which sites and territories deserve priority attention in an exploratory approach. Additional economic and technical data — grades, resources, CAPEX, OPEX, environmental constraints — would be needed to go beyond this first level of analysis.

This analysis therefore constitutes a first step in territorial screening. It does not replace a detailed geological or economic study, but it makes it possible to more effectively target the areas where a more in-depth analysis could be undertaken.

Bibliography / Sources

- USGS ScienceBase. Copper Arizona/Nevada project dataset. United States Geological Survey. https://www.sciencebase.gov/catalog/item/64dfc0efd34e5f6cd553c265. Accessed July 9, 2026.
- USGS. Mineral Resources Data System (MRDS). United States Geological Survey. https://mrdata.usgs.gov/mrds/. Accessed July 9, 2026.
- USGS. TopoMineSymbols Consolidated dataset (historical mining traces). United States Geological Survey. https://www.sciencebase.gov/catalog/item/64dfc0efd34e5f6cd553c265
- U.S. Census Bureau. Cartographic Boundary Files – Shapefile. https://www.census.gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html. Accessed July 9, 2026.
- OpenStreetMap contributors. Basemap used via QGIS (XYZ Tiles stream).

Methodological Appendix

This appendix gathers a few technical elements mentioned in the body of the report, intended for a reader wishing to reproduce or verify the processing carried out.

Copper Filtering Expression (QGIS)

```
"commod1" LIKE '%Copper%' OR "commod2" LIKE '%Copper%' OR "commod3" LIKE
'%Copper%'
```

Mining Context Expression (QGIS)

```
CASE WHEN "fid_count" >= 300 THEN 'Very strong' WHEN "fid_count" >= 100 THEN
'Strong' WHEN "fid_count" >= 30 THEN 'Medium' ELSE 'Low' END
```

Python Script

The copper_analysis.py script, placed in a scripts folder, reads the MRDS_copper_AZ_NV.gpkg and high_potential_10km_mining_points_count.gpkg layers. It relies on the pandas, geopandas, and matplotlib libraries to produce the CSV tables (outputs/tables folder) and PNG charts (outputs/figures folder) presented in this report.

File Organization

All processing is organized in a working folder named “Copper Exploration,” which notably brings together the following layers:

- MRDS_Arizona/ and MRDS_Nevada/ — raw MRDS data by state
- mrds-trim/ — preliminary extraction
- USGS_TopoMineSymbols_Consolidate... — historical mining traces
- cb_2018_us_state_500k/ — state administrative boundaries (Census Bureau)
- MRDS_Arizona_copper.gpkg and MRDS_Nevada_copper.gpkg — filtered copper sites by state
- MRDS_copper_AZ_NV.gpkg — Arizona/Nevada merge
- MRDS_copper_AZ_NV_clean.gpkg — cleaned database

- MRDS_copper_high_potential_AZ_NV.gpkg — sites classified as high potential
- mining_features_points_AZ_NV_clean.gpkg — cleaned mining traces
- buffer_10km_high_potential_copper_AZ_NV.gpkg — 10 km buffers around high-potential sites
- high_potential_10km_mining_points_count.gpkg — count of mining traces per buffer
- Projet_cuivre_AZ_NV_QGIS.qgz — complete QGIS project